VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS402

Compiler Design & Program Analysis - Comprehensive Syllabus Lecture Note

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Syllabus Focus: CS402 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Compiler Design & Program Analysis. Includes

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Lexical Analysis & Automata Generation

" Syllabus Scope: Role of lexical analyzer, tokens/patterns/lexemes, regular expressions to DFA Thompsons

Unit 2: Syntax Analysis & Top-Down Parsing

" Syllabus Scope: Context-free grammars, elimination of left recursion, left factoring, FIRST and FOLLOW

CS402: Compiler Design & Program Analysis - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Bottom-Up LR Parsing

" Shift-reduce parsing, handles, LR0 items, SLR1 parsing table, Canonical LR1 and LALR1 parsers, Yacc/Bison sp

Unit 4: Syntax-Directed Translation & Intermediate Code

" Syntax-Directed Definitions SDD, synthesized and inherited attributes, Three-Address Code TAC, quadruples, t

Unit 5: Code Optimization & Target Generation

" Basic blocks and flow graphs, local optimizations constant folding, dead code elimination, loop optimization

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Compute the FIRST and FOLLOW sets for a given grammar and construct its predictive LL1 parsing table.
2. Construct the set of LR0 items and the corresponding SLR1 parse table for an expression grammar, check
3. Translate an arithmetic expression with nested boolean conditions into Three-Address Code TAC quadrupl

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman - Compilers: Principles, Techniques, and To

Authorized NoteVault Educational Document " Verified for Registered University Students